

Quiz – Search

Instructions: This is a multiple-choice question section. For each question, choose **only one** correct answer.

* Enter your name and ID

* (1)Name

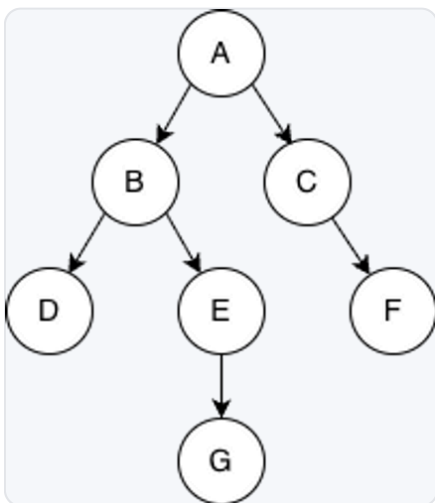
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* (2)Student ID

请输入

1. [单选] DFS (0.5分)

Start at A, goal is G. Using Depth-First Search, expand the successors of a node in the alphabetical order. What is the visitation order?



☐ A, B, C, D, E, F, G

☒ A, B, D, E, G

☐ A, C, F, B, E, G

☐ A, B, E, G, D

2. [单选] ID v.s. BFS (0.5分)

What is the main advantage of Iterative Deepening Search over Breadth-First Search?

- ☐ It is always faster
- ☒ It requires less memory
- ☐ It handles weighted graphs optimally
- ☐ It never revisits nodes

3. [单选] Heuristic (0.5分)

You have two admissible heuristics, h_1 and h_2 . Which of the following is NOT admissible?

- ☐ $\max(h_1, h_2)$
- ☐ $\min(h_1, h_2)$
- ☒ $h_1 + h_2$
- ☐ $(h_1 + h_2) / 2$

4. [单选] Informedness (0.5分)

Given that h_1 and h_2 are two different admissible heuristics, h_1 is more informed than h_2 . Which of the following statements is true?

- ☐ $\min(h_1, h_2)$ is more informed than h_1
- ☐ $h_1 + h_2$ is more informed than h_2
- ☒ $(h_1 + h_2) / 2$ is more informed than h_2
- ☐ $\max(h_1, h_2)$ is more informed than h_1

5. [单选] Pruning (0.5分)

In Depth-First Branch-and-Bound, you are searching for the cheapest path to a goal. You have a current best solution with cost C . At node n , the cost to reach n from the start is $c(n)$, and a heuristic estimate to the goal is $h(n)$. Under what condition can you safely prune node n ?

- ☒ $c(n) + h(n) \geq C$
- ☐ $c(n) + h(n) \leq C$
- ☐ $c(n) \leq C$
- ☐ $c(n) \geq C$